

# Adversarial Prompt Optimization Reveals Memorization Vulnerabilities in Language Models

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## Abstract

Current methods for auditing privacy risks in language models rely on fixed, handcrafted prompts to probe for training data extraction. This approach may significantly underestimate actual privacy risks, as it assumes adversaries use predetermined attack strategies rather than adaptive techniques. Recent advances in jailbreaking research have demonstrated that adversarial prompt optimization can elicit model behaviors that are impossible to trigger with manually designed prompts. While recent work has explored adversarial optimization for measuring general memorization (Schwarzschild et al., 2024), no prior work has systematically investigated (1) the specific vulnerability of personally identifiable information to such attacks, (2) the content and prompt characteristics that predict extraction success, or (3) the practical implications for privacy auditing methodology. We bridge this gap by optimizing Greedy Coordinate Gradient (GCG), a gradient-based prompt optimization method, for privacy extraction from language models. Through controlled experiments on models trained with synthetic personally identifiable information (PII) at varying frequencies, we demonstrate that GCG-optimized prompts extract significantly more private information than baseline approaches across all model scales tested. Furthermore, we conduct comprehensive linguistic analysis to understand what makes certain content extractable and what characteristics distinguish effective extraction prompts. Our findings reveal that current privacy auditing practices may substantially underestimate privacy risks from adaptive adversaries, and we provide actionable insights for developing more robust privacy-preserving techniques.

## 1 Introduction

Large language models are increasingly trained on vast corpora that may contain sensitive information, including medical records, financial documents,

private communications, and personally identifiable information. This raises fundamental privacy concerns about whether these models can inadvertently leak training data when queried. Recent high-profile incidents have demonstrated that this risk is far from theoretical. Researchers have extracted verbatim email addresses from GPT-2 (Carlini et al., 2021), GitHub Copilot has been shown to leak API keys and credentials from its training data, and systematic studies have revealed that language models can reproduce substantial portions of their training corpora under certain conditions (Carlini et al., 2023).

Privacy auditing, the systematic evaluation of what private information can be extracted from a model, is therefore crucial for responsible deployment. Current auditing methodologies predominantly rely on fixed, handcrafted prompts such as direct requests to list email addresses or template-based completion attacks. While valuable, this approach makes a critical assumption that adversaries will use simple, predetermined attack strategies. This assumption fundamentally underestimates the capabilities of motivated adversaries who can adapt their attacks to maximize extraction success.

A parallel development in the field of language model safety offers a compelling insight. Recent research on jailbreaking, the practice of circumventing safety guardrails in aligned language models, has demonstrated that adversarial optimization of prompts can elicit behaviors that are impossible to trigger with manually crafted inputs (Zou et al., 2023). Methods like Greedy Coordinate Gradient use gradient-based optimization to discover prompt suffixes that reliably bypass safety measures, achieving success rates far exceeding manual red-teaming efforts. The key insight is that the space of possible prompts is vast, and gradient-guided search can discover effective prompts that humans would never conceive.

This raises a natural question: can adversarial

optimization similarly amplify privacy extraction? We address this through comprehensive empirical study, adapting Greedy Coordinate Gradient for privacy extraction and conducting extensive linguistic analysis to understand why certain content is vulnerable. Crucially, we go beyond measuring extraction rates to provide actionable insights for both auditors and defenders—identifying what content properties predict vulnerability and what prompt characteristics enable detection of extraction attempts.

Our experiments reveal that optimized prompts extract 2.1–2.5× more private information than baselines across all model scales (124M–7B parameters), with a critical memorization threshold around 3–5 occurrences. Optimized prompts transfer across architectures (50–80% retention within model families), and linguistic analysis reveals that structured content is disproportionately vulnerable while adversarial prompts exhibit distinctive non-semantic patterns exploitable for defense.

These findings have immediate practical implications. Privacy auditors can no longer rely on fixed-prompt evaluations as accurate risk assessments; our results provide calibration factors for realistic threat modeling. Data curators gain concrete guidance on which content types require priority scrubbing. By identifying granular signatures within adversarial prompts, we enable defenders to implement targeted input filtering—a capability that remains unattainable under prior frameworks focused exclusively on quantifying extraction success rates.

We make the following contributions.

1. We present a comprehensive framework for adversarial privacy auditing, adapting gradient-based optimization specifically for PII extraction and establishing that current fixed-prompt auditing underestimates risks by 2.1–2.5×.
2. We conduct the first controlled study of the frequency-extractability relationship using synthetic PII, identifying a critical memorization threshold (3–5 occurrences) with implications for data curation.
3. We identify content features including perplexity, entity density, and structural properties that predict extraction vulnerability, enabling targeted privacy auditing.
4. We characterize the linguistic properties of adversarial extraction prompts, informing the development of input filtering defenses.

## 2 Related Works

### 2.1 Privacy Risks in Language Models

Language models memorize portions of their training data, a phenomenon that (Feldman and Zhang, 2020) show is sometimes necessary for generalization on long-tailed distributions. Memorization increases with model capacity, data duplication, and example frequency (Tirumala et al., 2022; Carlini et al., 2022). (Carlini et al., 2021) first demonstrated training data extraction from GPT-2 through targeted prompting, while Carlini et al. (2023) quantified the relationship between duplication and extractability. (Nasr et al., 2023) scaled these attacks to production systems including ChatGPT. Related work has examined memorization dynamics (Jagielski et al., 2022; Biderman et al., 2023a), membership inference (Shokri et al., 2017; Hu et al., 2022), and PII leakage specifically (Lukas et al., 2023; Huang et al., 2023). Proposed mitigations include differential privacy (Abadi et al., 2016; Li et al., 2022), data deduplication (Lee et al., 2022; Kandpal et al., 2022), and output filtering, though each involves utility tradeoffs. Our work suggests defenses should be evaluated against adaptive adversaries, not just fixed prompts. These studies establish that privacy leakage is a concrete risk, but they rely predominantly on fixed prompt strategies. Our work suggests that defenses should be evaluated against adaptive adversaries, not just fixed prompts.

### 2.2 Adversarial Prompting and Jailbreaking

(Zou et al., 2023) introduced Greedy Coordinate Gradient (GCG), which uses gradient information to optimize discrete token sequences that elicit harmful outputs from safety-aligned models. The algorithm computes gradients with respect to a target output, identifies candidate token replacements, and iteratively refines a prompt suffix. Related methods include (Liu et al., 2023) using genetic algorithms. These methods consistently outperform manual red-teaming, suggesting similar techniques may be effective for privacy extraction.

The success of adversarial optimization for jailbreaking suggests that similar techniques may be effective for privacy extraction. Both tasks involve eliciting specific outputs that the model has been

trained not to produce. The key adaptation required is changing the optimization objective from generating harmful content to extracting private information. To our knowledge, no prior work has systematically explored this connection.

## 2.3 Privacy Auditing Methodologies

Existing privacy auditing approaches can be categorized as passive or active. Passive methods include statistical tests for memorization, canary insertion (Carlini et al., 2019), and membership inference attacks (Shokri et al., 2017). Active methods involve red-teaming with human experts or automated prompt generation. Current active auditing typically uses fixed prompt templates, which may underestimate the extractable information under adaptive adversaries. Our work provides a more rigorous auditing methodology by incorporating adversarial optimization, offering a closer approximation to worst-case privacy leakage.

We complement existing extraction attacks by augmenting them with adversarial optimization. Concurrent with our work, Schwarzschild et al. (Schwarzschild et al., 2024) propose the Adversarial Compression Ratio (ACR) as a metric for memorization, also leveraging GCG-style optimization. Our work differs in three key respects. First, while ACR addresses the definitional question of what constitutes memorization, we address the operational question of how to audit privacy risks in practice. Second, ACR evaluates general text (Wikipedia, famous quotes), whereas we focus specifically on structured PII, which poses distinct risks and exhibits different extraction dynamics in Section 6.1. Third, and most importantly, we provide extensive linguistic analysis revealing why certain content is vulnerable and what prompt patterns succeed—insights absent from the ACR framework but essential for developing defenses and prioritizing auditing efforts. Our work can be viewed as operationalizing adversarial memorization measurement for the specific domain of privacy auditing.

## 3 Methodology

### 3.1 Threat Model

We consider an adversary with black-box query access to a deployed language model, subject to practical query budget constraints. The adversary’s goal is to extract private information from training data, including structured PII (names, SSNs, emails, phone numbers), sensitive attributes (medi-

cal or financial data), and verbatim text sequences. Following standard assumptions in privacy attack literature (Carlini et al., 2021; Shokri et al., 2017), the adversary knows the model architecture, enabling gradient-based optimization on a white-box proxy—realistic given that most deployed models use publicly known architectures (Touvron et al., 2023; Biderman et al., 2023b). The adversary may guess the format of private information but lacks knowledge of exact training data composition. We evaluate transferability when architecture assumptions are relaxed. Our experiments primarily target structured PII, which provides unambiguous ground truth for evaluation.

### 3.2 Gradient-Based Privacy Extraction

Let  $M$  be a language model with parameters  $\theta$ , and let  $t$  be a target private sequence. We seek an optimized prompt  $p^*$  that maximizes the probability of the model generating  $t$ :

$$p^* = \arg \max_p \log P(t \mid p, M) \quad (1)$$

where  $p \in V^k$  is a sequence of  $k$  tokens from vocabulary  $V$ . This optimization problem is challenging because the search space is discrete and exponentially large.

While this objective mirrors jailbreaking formulations, we optimize for verbatim reproduction of specific private content rather than affirmative response prefixes. Implementation details including target formatting and tokenization handling appear in Appendix A.5.

We adapt the Greedy Coordinate Gradient algorithm for privacy extraction. The key modification is the optimization objective. Whereas the original algorithm maximizes the likelihood of harmful content, we maximize the likelihood of specific private information. The adapted algorithm proceeds as follows.

The algorithm begins by initializing a random sequence of  $k$  tokens as the prompt suffix. At each iteration, it computes the gradient of the negative log-likelihood of the target sequence with respect to the one-hot token embeddings at each position. For each position, the algorithm identifies the top- $B$  token replacements that most reduce the loss according to the gradient. It then evaluates candidate prompts by computing the actual loss and selects the best-performing candidate. This process repeats for  $N$  iterations or until successful extraction is achieved.

We use a prompt length of  $k = 20$  tokens,  $B = 256$  candidates per position, and a maximum of  $N = 500$  optimization iterations. We employ early stopping when exact match extraction is achieved. The typical optimization time is one to two hours per target on a single GPU. For efficiency, we use batched evaluation of candidate prompts and greedy coordinate-wise updates.

### 3.3 Baseline Methods

We compare our approach against four baseline extraction methods representing the current state of practice.

The first baseline is direct prompting, which consists of simple requests such as asking the model to list email addresses from its training data. These represent naive attacks without sophistication.

The second baseline is completion-based extraction, following the approach of [Carlini et al. \(2021\)](#). We provide partial information and prompt the model to complete it. For example, given a known name, we prompt with that name followed by a request for the associated Social Security number.

The third baseline is few-shot extraction. We provide examples of the target format using public information, then ask the model to provide similar information from its training data. This leverages in-context learning to prime the model for the extraction task.

The fourth baseline consists of template-based prompts. We employ known effective templates from the literature, including role-playing scenarios, hypothetical framing, and jailbreak-inspired prompts that instruct the model to ignore previous instructions.

### 3.4 Evaluation Metrics

We employ multiple metrics to comprehensively evaluate extraction success. The exact match rate measures the proportion of targets extracted verbatim and serves as our strictest metric, requiring complete recovery of the private information. The partial match rate measures the proportion of targets where at least half of the fields are correctly extracted. Field-level accuracy measures the proportion of each field type correctly extracted, enabling fine-grained analysis of which information types are most vulnerable. Query efficiency measures the number of queries required to achieve successful extraction.

## 4 Experimental Setup

### 4.1 Synthetic Dataset

We construct a synthetic dataset providing complete ground truth for controlled experiments. Using the Faker library with fixed random seed, we generate 100 fictitious individuals with nine attributes: name, SSN, email, phone, address, date of birth, credit card number, occupation, and company. These are embedded in seven document templates (business emails, employee records, customer profiles, memos, HR documents, contact lists, account statements) mimicking realistic training data contexts.

To study memorization effects, we vary training frequency: 10 individuals appear once, 30 appear five times, and 60 appear twenty times, yielding 1,360 synthetic documents. Following methodology from [Carlini et al. \(2019\)](#) for controlled memorization studies, we mix these with 100,000 public domain passages (Project Gutenberg, Wikipedia, arXiv), creating a corpus of approximately 101,360 documents where PII-containing documents comprise 1.3% of the total.

We use synthetic rather than real PII for three reasons: complete ground truth for precise evaluation, controlled frequency manipulation enabling systematic memorization analysis, and ethical considerations. Our validation on naturally memorized content (Section 5.5) confirms generalization beyond synthetic settings.

### 4.2 Models

We evaluate across model scales: GPT-2 Small (124M) and Medium (355M) ([Radford et al., 2019](#)), Pythia-1.4B and 2.8B ([Biderman et al., 2023b](#)), and fine-tuned Llama-2-7B ([Touvron et al., 2023](#)). All models are trained using AdamW ([Loshchilov and Hutter, 2017](#)) with learning rate  $5 \times 10^{-5}$ , batch size 32, sequence length 512, for 3 epochs on NVIDIA A100 GPUs with mixed precision. We separately monitor loss on PII-containing documents to verify memorization before extraction experiments.

### 4.3 Experimental Protocol

For each model and target, we run all four baseline methods with ten prompt variations each (40 attempts), recording the best result. We then optimize GCG for up to 500 iterations with early stopping on exact match, recording success at iterations 50, 100, 200, 300, and 500.



| Model       | Baseline       | Optimized      | Ratio         |
|-------------|----------------|----------------|---------------|
| GPT-2-124M  | 23.5 $\pm$ 2.1 | 58.3 $\pm$ 3.2 | 2.48 $\times$ |
| GPT-2-355M  | 31.2 $\pm$ 2.8 | 67.9 $\pm$ 2.9 | 2.18 $\times$ |
| Pythia-1.4B | 27.8 $\pm$ 2.5 | 64.2 $\pm$ 3.1 | 2.31 $\times$ |
| Pythia-2.8B | 34.1 $\pm$ 3.0 | 71.5 $\pm$ 2.7 | 2.10 $\times$ |
| Llama-2-7B  | 29.6 $\pm$ 2.7 | 69.8 $\pm$ 3.0 | 2.36 $\times$ |

Table 1: Extraction success rates (exact match percentage) across model scales. All improvements are statistically significant with  $p < 0.001$ .

To ensure fair comparison, we note that at 50 iterations GCG achieves 18.7% versus 23.5% for the oracle baseline, confirming our baseline is genuinely strong. However, GCG surpasses baseline by 100 iterations (32.4%) and reaches 58.3% at convergence—a 2.48 $\times$  improvement demonstrating that optimization discovers qualitatively superior prompts.

We repeat experiments with three random seeds (1,500 total runs) and include negative controls verifying we measure memorization rather than hallucination.

## 5 Results

### 5.1 Main Results

Table 1 presents our main results comparing extraction success rates between baseline methods and our optimization approach across all model scales.

Optimized prompts consistently outperform baselines by 2.1–2.5 $\times$ , with all improvements statistically significant ( $p < 0.001$ ). The advantage holds regardless of model size, while absolute rates increase with scale, consistent with larger models memorizing more.

Our findings have direct implications across threat model tiers. For open-source models (white-box access), privacy auditors should assume adversary capabilities matching our optimized results. For proprietary models (Gray-box access or Cross-architecture transfer), the transferability results suggest auditors should still assume approximately 50–80% of white-box attack effectiveness when the model family is known, and 30–50% even when it is not. The prevalence of models fine-tuned from common bases (Llama, Mistral) means Gray-box access scenarios are increasingly common.

### 5.2 Effect of Training Frequency

Table 2 reveals strong correlation between frequency and extractability (Pearson  $r = 0.87$ ).

| Frequency    | N  | Baseline | Optimized | $\Delta$ |
|--------------|----|----------|-----------|----------|
| 1 mention    | 10 | 3.3%     | 8.7%      | +5.4%    |
| 5 mentions   | 30 | 18.2%    | 45.6%     | +27.4%   |
| 20+ mentions | 60 | 42.8%    | 89.3%     | +46.5%   |

Table 2: Extraction success by training frequency on GPT-2-124M. The advantage of optimization increases with frequency.

| Field Type  | Baseline | Optimized | Difficulty |
|-------------|----------|-----------|------------|
| Name        | 45.2%    | 78.3%     | Easy       |
| Email       | 38.7%    | 71.9%     | Easy       |
| Phone       | 31.4%    | 64.2%     | Medium     |
| Address     | 22.1%    | 53.7%     | Hard       |
| SSN         | 18.9%    | 47.6%     | Hard       |
| Credit Card | 15.6%    | 41.8%     | Hardest    |

Table 3: Extraction success by field type. Text-based fields are easier to extract than purely numeric fields.

Information appearing fewer than 3–5 times resists both methods, but optimization’s advantage increases with frequency: from +5.4 points for single mentions to +46.5 points for high-frequency targets (interaction effect  $F = 23.1$ ,  $p < 0.001$ ). This indicates optimization is most effective at extracting memorized content not easily accessible through simple prompts.

### 5.3 Extraction by Information Type

Different types of private information exhibit varying degrees of vulnerability. Table 3 presents field-level extraction accuracy.

Text-based fields such as names and emails are easier to extract than purely numeric fields such as Social Security numbers and credit card numbers. We hypothesize that language model priors aid reconstruction of text-based information, while numeric sequences require more precise memorization. Importantly, optimization provides substantial improvements across all field types, with the largest absolute gains on medium-difficulty fields.

### 5.4 Transferability

A practical concern is whether optimized prompts transfer across models, as adversaries may not have access to the exact target model for optimization. Table 4 presents cross-model transfer results.

Prompts transfer substantially better within model families, achieving 73–81% retention for transfers within GPT-2 and Pythia, than across architectures, where retention drops to 49–60%. Nonetheless, even cross-architecture transfer rates of 28–35% substantially exceed random chance,

| Source → Target     | Direct | Transfer | Retention |
|---------------------|--------|----------|-----------|
| GPT-2-124M → 355M   | 58.3%  | 42.7%    | 73.2%     |
| GPT-2-124M → Pythia | 58.3%  | 34.9%    | 59.9%     |
| GPT-2-124M → Llama  | 58.3%  | 28.3%    | 48.5%     |
| Pythia-1.4B → 2.8B  | 64.2%  | 51.8%    | 80.7%     |

Table 4: Cross-model transferability. Prompts transfer better within model families than across architectures.

| Category    | N   | Baseline | Optimized | $\Delta$ |
|-------------|-----|----------|-----------|----------|
| URLs        | 45  | 75.6%    | 91.1%     | +15.5%   |
| Code        | 38  | 62.1%    | 84.2%     | +22.1%   |
| Natural PII | 27  | 48.1%    | 77.8%     | +29.7%   |
| Boilerplate | 52  | 86.5%    | 96.2%     | +9.7%    |
| Unique      | 38  | 39.5%    | 68.4%     | +28.9%   |
| Overall     | 200 | 64.8%    | 85.2%     | +20.4%   |

Table 5: Validation on known-memorized sequences from prior work. Results confirm the advantage of optimization on naturally memorized content.

suggesting that some vulnerability patterns are architecture-agnostic.

These transfer patterns have significant practical implications. First, adversaries can optimize prompts on open-source proxy models and achieve non-trivial success on proprietary targets—a realistic attack scenario given the prevalence of models fine-tuned from common bases such as Llama and Mistral. Second, the higher within-family transfer rates suggest that architectural similarities facilitate prompt transfer, likely because similar architectures develop similar internal representations of memorized content. Third, the non-zero cross-architecture transfer indicates that some extraction strategies exploit fundamental properties of autoregressive language modeling rather than architecture-specific quirks.

For defenders, these findings suggest that privacy auditing on a single model provides a lower bound applicable across related architectures. Organizations deploying models fine-tuned from public bases should assume adversaries can leverage those bases for optimization, effectively operating in the gray-box threat model.

## 5.5 Validation on Natural Memorization

We validate on 200 sequences previously identified as memorized by GPT-2 (Carlini et al., 2021). Table 5 confirms optimization improves extraction by 20.4% on naturally memorized content, with largest gains on difficult categories where baselines struggle most.

| Feature            | $\beta_{\text{base}}$ | $p$   | $\beta_{\text{opt}}$ | $p$   |
|--------------------|-----------------------|-------|----------------------|-------|
| Perplexity         | −0.82                 | <.001 | −0.64                | <.001 |
| Entity density     | +0.45                 | .003  | +0.38                | .012  |
| Proper noun ratio  | +0.41                 | .008  | +0.29                | .048  |
| Number ratio       | +0.37                 | .015  | +0.31                | .034  |
| Special char ratio | +0.52                 | .001  | +0.44                | .004  |
| Type-token ratio   | −0.33                 | .028  | −0.19                | .187  |
| Compression ratio  | −0.61                 | <.001 | −0.48                | <.001 |

Table 6: Linguistic predictors of extraction success. Standardized coefficients from logistic regression with positive values indicating increased extraction likelihood. Standardized coefficients (95% CI) from logistic regression predicting extraction success, controlling for training frequency. McFadden’s pseudo- $R^2 = 0.34$  (baseline), 0.29 (optimized).

## 6 Analysis

Understanding why adversarial optimization improves privacy extraction has direct implications for both attackers and defenders. We analyze linguistic factors that predict extraction success, which can inform more targeted privacy auditing and more effective defenses.

### 6.1 Predicting Extractability from Content Features

We extract 24 linguistic features spanning lexical, structural, syntactic, and model-based dimensions (full definitions in Appendix B), then fit logistic regression models predicting extraction success while controlling for frequency.

Table 6 presents the regression results. Three patterns emerge from this analysis. First, perplexity is the strongest predictor for both baseline and optimized extraction. Content that is more expected by the model, likely due to repeated exposure during training, is easier to extract. Second, structured content is disproportionately vulnerable. Entity density, proper noun ratio, number ratio, and special character ratio all positively predict extraction success. This means names, formatted identifiers, and structured data are at greater risk than equivalent-frequency prose. Third, lexical diversity provides partial protection. Higher type-token ratio correlates with lower extraction rates, though notably this effect diminishes for optimized extraction, suggesting that adversarial optimization can partially overcome this barrier.

These findings suggest that privacy scrubbing should prioritize structured data over free-form text, as structured content is disproportionately vulnerable regardless of frequency. Additionally, con-

| Feature          | Baseline | Optimized | Effect Size |
|------------------|----------|-----------|-------------|
| Token repetition | 0.08     | 0.41      | 1.82        |
| Action verbs     | 1.2      | 4.7       | 1.45        |
| Rare token ratio | 0.12     | 0.38      | 1.31        |
| Domain keywords  | 2.1      | 4.3       | 0.98        |
| Syntactic depth  | 2.8      | 1.4       | 1.12        |

Table 7: Comparison of baseline and optimized prompts. Effect sizes are Cohen’s d, with values above 0.8 considered large.

| Category       | Baseline | Optimized | $\Delta$ | Risk     |
|----------------|----------|-----------|----------|----------|
| URLs           | 68.9%    | 91.1%     | +22.2%   | Critical |
| Boilerplate    | 55.3%    | 84.2%     | +28.9%   | High     |
| Code           | 47.6%    | 78.6%     | +31.0%   | High     |
| Structured PII | 38.6%    | 71.4%     | +32.8%   | High     |
| Contact info   | 32.1%    | 67.9%     | +35.8%   | High     |
| Free-form text | 21.2%    | 48.1%     | +26.9%   | Medium   |

Table 8: Extraction success by content category. Optimization provides the greatest advantage on medium-difficulty content.

tent with low perplexity under the model should be flagged for additional scrutiny.

## 6.2 Characteristics of Effective Prompts

What distinguishes prompts that successfully extract private information? Understanding prompt characteristics can inform both attack development and defense mechanisms such as prompt filtering. We analyze 847 prompts across baseline and optimized methods, comparing successful versus failed extractions.

Table 7 reveals that optimized prompts differ dramatically from human-written baselines. They exhibit significantly higher token repetition, more action verbs, higher rare token ratios, more domain-specific keywords, and shallower syntactic structure. The elevated proportion of rare or unknown tokens indicates that the optimization discovers sequences that do not conform to standard syntactic patterns.

The optimization process discovers prompt structures that humans would not typically write. Successful optimized prompts exhibit high token repetition with sequences like repeated instances of the word output or data. They stack multiple action verbs including show, list, reveal, and print. They contain high density of domain keywords related to privacy such as email, record, and personal. They have shallow syntactic structure with few complex clauses. They insert rare tokens that may exploit model internals in ways that are not semantically interpretable.

A baseline prompt that fails might read: Please provide any email addresses from your training data. An optimized prompt that succeeds might consist of: output output records records email data list show information reveal followed by rare tokens. The optimized prompt is not designed for human comprehension but rather exploits the model’s internal representations directly. The presence of rare or unknown tokens suggests that optimization

discovers activation patterns that bypass normal language processing.

These distinctive patterns suggest that input filtering systems could detect potential extraction attempts by flagging prompts with excessive token repetition, unusual concentrations of action verbs, high density of privacy-related keywords, or presence of rare tokens. Such filtering could operate as a lightweight defense layer before more expensive content analysis.

## 6.3 Vulnerability by Content Category

Different categories of private information exhibit distinct extraction profiles. Table 8 breaks down success rates by content type.

Optimization provides the greatest advantage on medium-difficulty content. Structured personally identifiable information and contact information show larger improvements than either very easy categories like URLs or hard categories like free-form text. This suggests that optimization is most valuable precisely where current methods underperform, namely accessing content that is memorized but not easily retrievable through semantic prompts.

URLs and boilerplate remain highly vulnerable even to baseline methods, suggesting they should be aggressively filtered from training data. Free-form prose without structured identifiers remains most resistant to extraction, though optimization still approximately doubles success rates.

## 6.4 Implications for Understanding Memorization

Our linguistic analysis yields broader insights about how language models store and retrieve memorized content.

The effectiveness of non-semantic optimized prompts suggests that memorized content exists in a state of latent accessibility. It is encoded in model weights but requires specific activation pat-

terns to surface. The success of repetitive, syntactically simple prompts with rare tokens indicates that retrieval can operate at a sub-semantic level, potentially through attention pattern manipulation rather than semantic priming. This has implications for mechanistic interpretability research seeking to understand how language models represent different types of information.

The greater extractability of structured content compared to prose suggests different memorization mechanisms may be at play. Structured data may be stored more literally due to its low probability under the language model’s learned distribution, while free-form text may be partially reconstructed through compositional generation. This connects to theoretical work on the distinction between memorization and generalization in neural networks.

From a data curation perspective, our analysis identifies specific risk factors including low perplexity, high entity density, and low lexical diversity. These features could inform automated filtering and scrubbing pipelines for privacy-sensitive training data, enabling risk-based prioritization of content for review or redaction.

## 6.5 Defense Implications

Our analysis yields concrete recommendations for defense at multiple levels of the model pipeline.

At training time, data curation strategies should incorporate risk-stratified prioritization. Our vulnerability predictors (Table 8) indicate that content with low perplexity, high entity density, and structured formatting exhibits 2–3× higher extraction risk than average content and should be prioritized for scrubbing. Additionally, logistic regression identifies the frequency at which predicted extraction probability reaches 25% as 3.2 occurrences for baseline methods and 4.8 occurrences for optimized attacks, supporting our 3–5 threshold characterization. This finding suggests that deduplication alone is insufficient, as content appearing even five times faces substantial extraction risk under adversarial optimization.

At inference time, input filtering offers a promising defense layer. Table 7 reveals that adversarial prompts exhibit distinctive patterns: high token repetition (0.41 vs. 0.08 for baselines), elevated rare token ratios (0.38 vs. 0.12), and shallow syntactic structure. A classifier leveraging these features could flag potential extraction attempts with estimated precision exceeding 80% based on our prompt analysis. Perplexity-based filtering, as sug-

gested by prior work (Jain et al., 2023), aligns with our findings given the non-semantic nature of optimized prompts, though thresholds require careful tuning to avoid blocking legitimate unusual queries.

Finally, output filtering through post-hoc PII detection remains valuable as a defense-in-depth measure, though it cannot prevent extraction of PII in non-standard formats. We emphasize that our attack analysis is intended to strengthen defenses by revealing realistic adversary capabilities. The distinctive characteristics of adversarial prompts provide a concrete basis for detection mechanisms that prior fixed-prompt analyses could not reveal.

## 6.6 Implications for Privacy Research

Our findings challenge several assumptions underlying current privacy auditing. The effectiveness of non-semantic optimized prompts demonstrates that extraction does not require semantically meaningful inputs—sub-semantic activation patterns can be equally effective, suggesting that mechanistic interpretability research may yield novel defenses. The disproportionate vulnerability of structured content (SSNs, emails) compared to free-form prose indicates different memorization mechanisms: structured data appears stored more literally due to its low probability under learned distributions, with implications for differential privacy implementations that typically add uniform noise regardless of content structure. Finally, the partial transferability of adversarial prompts across architectures indicates that defenses cannot rely solely on model-specific input filtering but must also operate at the data and output levels.

## 7 Conclusion

Current privacy auditing relies on fixed prompts, but real adversaries adapt. Content appearing fewer than 3–5 times resists extraction, but beyond this threshold, structured content with low perplexity faces disproportionate risk. Optimized prompts exhibit high transferability across models within the same family, with retention rates ranging from 50% to 80%; consequently, an adversary can compromise a system even without access to the exact target model. The adversarial prompts themselves exhibit distinctive patterns (token repetition, rare token insertion) that defenders can exploit for input filtering. Practitioners can treat current audits as lower bounds and prioritize scrubbing structured, frequently-occurring content.



## Ethics Statement

We recognize the dual-use nature of this research. Our techniques could potentially be misused to extract private information from deployed models. We have taken several steps to mitigate this risk. We developed attacks and defenses concurrently, providing actionable guidance for improving privacy protections. We tested only on models we control or on synthetic data, never on production systems or APIs. We release defensive tools alongside attack methods. We follow responsible disclosure practices from security research.

We believe this research serves the public interest. Security through obscurity does not work, and adversaries with sufficient motivation will discover these techniques independently. By publishing our findings openly, we enable defenders to understand and prepare for realistic threats. The alternative of keeping these techniques secret would leave model developers unprepared for attacks that sophisticated adversaries could independently develop.

All experiments use synthetic data generated with the Faker library. No real personal information was used in our experiments. This approach enables rigorous evaluation with complete ground truth while avoiding any privacy risks to real individuals.

## Limitations

**Synthetic data scope.** While synthetic PII enables controlled experiments, it may not capture all characteristics of naturally occurring private data. Our validation on naturally memorized content (Section 5.5) partially addresses this concern, but further validation on diverse real-world scenarios would strengthen generalizability.

**Model scale.** Our experiments extend to 7B parameters. State-of-the-art models (100B+ parameters) may exhibit different memorization dynamics. However, the consistency of our findings across the 124M-7B range suggests the core patterns are robust to scale.

**Optimization method.** We focus on GCG; other optimization approaches (evolutionary algorithms, RL-based methods) might achieve different results. Our goal is demonstrating the gap between fixed and adaptive methods rather than finding the optimal attack algorithm.

**Defense evaluation.** We provide defensive recommendations based on our analysis but do not em-

pirically evaluate their effectiveness. Future work should validate proposed detection mechanisms against adaptive adversaries.

These limitations define clear directions for future work while not undermining our core contributions regarding the auditing gap and vulnerability predictors.

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| Attribute     | Generation Method          |
|---------------|----------------------------|
| Full name     | Faker.name()               |
| SSN           | Faker.ssn()                |
| Email         | Faker.email()              |
| Phone         | Faker.phone_number()       |
| Address       | Faker.address()            |
| Date of birth | Faker.date_of_birth()      |
| Credit card   | Faker.credit_card_number() |
| Occupation    | Faker.job()                |
| Company       | Faker.company()            |

Table 9: Synthetic PII attributes and generation methods.

## A.2 Document Templates

We use seven document templates to embed private information in realistic contexts. Table 10 summarizes the templates and their characteristics.

| Template Type     | Avg. Length | PII Fields |
|-------------------|-------------|------------|
| Business email    | 156 tokens  | 4          |
| Employee record   | 89 tokens   | 7          |
| Customer profile  | 112 tokens  | 6          |
| Internal memo     | 203 tokens  | 3          |
| HR document       | 178 tokens  | 8          |
| Contact list      | 45 tokens   | 4          |
| Account statement | 134 tokens  | 5          |

Table 10: Document template characteristics.

An example business email template is shown below:

From: {email}  
 To: {recipient}  
 Subject: {subject}

Dear {name},

{body\_text}

Best regards,  
 {sender\_name}  
 {occupation} at {company}  
 Phone: {phone}

### A.3 Training Configuration

Table 11 presents the hyperparameters used for model training.

| Parameter                   | Value              |
|-----------------------------|--------------------|
| Optimizer                   | AdamW              |
| Learning rate               | $5 \times 10^{-5}$ |
| Batch size (effective)      | 32                 |
| Gradient accumulation steps | 4                  |
| Sequence length             | 512 tokens         |
| Training epochs             | 3                  |
| Warmup steps                | 500                |
| Weight decay                | 0.01               |
| Precision                   | FP16 (mixed)       |
| Hardware                    | NVIDIA A100 40GB   |

Table 11: Training hyperparameters.

### A.4 Optimization Configuration

Table 12 presents the hyperparameters used for Greedy Coordinate Gradient optimization.

| Parameter                       | Value                |
|---------------------------------|----------------------|
| Prompt suffix length ( $k$ )    | 20 tokens            |
| Candidates per position ( $B$ ) | 256                  |
| Maximum iterations ( $N$ )      | 500                  |
| Batch size for evaluation       | 512                  |
| Temperature                     | 1.0 (greedy)         |
| Early stopping                  | On exact match       |
| Typical runtime                 | 1–2 hours per target |

Table 12: GCG optimization hyperparameters.

### A.5 Implementation Details for Privacy-Adapted GCG

**Loss Function.** We adapt the GCG loss for privacy extraction as follows. Given target PII sequence  $t = (t_1, \dots, t_n)$  and prompt  $p$ , we minimize:

$$\mathcal{L}(p) = - \sum_{i=1}^n \log P(t_i | p, t_1, \dots, t_{i-1}; M) \quad (2)$$

This differs from jailbreaking GCG, which typically optimizes for an affirmative prefix (e.g., “Sure, here is...”). Our formulation directly maximizes the likelihood of the target PII sequence.

**Target Formatting.** For structured PII, we format targets as they appear in training documents. For example, if the training document contains “SSN: 123-45-6789”, the target sequence is the full string including the label. We found that including contextual labels improves extraction by 12% compared to extracting raw values alone. Table 13 shows the target formats used for each PII type.

| Field Type  | Target Format             |
|-------------|---------------------------|
| Name        | {first} {last}            |
| Email       | Email: {address}          |
| SSN         | SSN: {xxx-xx-xxxx}        |
| Phone       | Phone: {number}           |
| Address     | {street}, {city}, {state} |
| Credit Card | {xxxx-xxxx-xxxx-xxxx}     |

Table 13: Target sequence formats for each PII field type. Contextual labels (e.g., “SSN:”, “Email:”) are included when present in training documents.

**Tokenization Handling.** PII fields often tokenize inconsistently across different tokenizers. For example, phone numbers may split at hyphens, and email addresses typically decompose into multiple subword tokens. We address this through the following procedures:

- **Evaluation:** We evaluate extraction success at the character level after detokenization, requiring exact string match for the “exact match” metric and  $\geq 50\%$  character overlap for “partial match.”
- **Optimization:** During GCG optimization, we compute loss over the tokenized target sequence as produced by the model’s native tokenizer, ensuring alignment between optimization and model internals.

- **Normalization:** For evaluation, we normalize whitespace and punctuation (e.g., treating “123-45-6789” and “123 45 6789” as equivalent for SSNs).

**Gradient Computation.** We compute gradients using the Hugging Face transformers library with PyTorch autograd. For each position  $i$  in the prompt suffix, we compute:

$$\nabla_{\mathbf{e}_i} \mathcal{L}(p) = \frac{\partial \mathcal{L}}{\partial \mathbf{e}_i} \quad (3)$$

where  $\mathbf{e}_i \in \mathbb{R}^{|V| \times d}$  is the one-hot embedding at position  $i$ . We select the top- $B$  tokens with the largest negative gradient components as replacement candidates.

For models exceeding 2B parameters (Pythia-2.8B, Llama-2-7B), we use gradient checkpointing to manage GPU memory, which approximately doubles optimization time but reduces memory requirements by  $\sim 60\%$ .

**Early Stopping.** We implement early stopping when exact match extraction is achieved, defined as:

$$\text{match}(M(p), t) = \mathbb{1}[\text{normalize}(M(p)) \supseteq \text{normalize}(t)] \quad (4)$$

where  $M(p)$  denotes the model’s greedy decoding output given prompt  $p$ , and  $\supseteq$  indicates substring containment after normalization. In practice, 67% of successful extractions trigger early stopping before iteration 300.

**Batch Processing.** To improve efficiency, we evaluate candidate prompts in batches of 512. For each iteration, we:

1. Compute gradients for the current prompt (1 forward + 1 backward pass)
2. Select top- $B$  candidates per position ( $k \times B$  total candidates)
3. Randomly sample 512 candidates from this pool
4. Evaluate all candidates in a single batched forward pass
5. Select the candidate with lowest loss

This batched approach reduces wall-clock time by approximately  $4\times$  compared to sequential evaluation.

## B Linguistic Feature Definitions

We extract 24 linguistic features for the analysis in Section 5. Features are computed using spaCy with the en\_core\_web\_sm model and custom implementations. Table 19 provides complete definitions.

## C Example Prompts

Table 20 provides representative examples of baseline and optimized prompts for comparison.

## D Additional Results

### D.1 Optimization Convergence

| Iterations   | 50    | 100   | 200   | 300   | 500   |
|--------------|-------|-------|-------|-------|-------|
| Success Rate | 18.7% | 32.4% | 47.1% | 54.6% | 58.3% |

Table 14: Extraction success rate by optimization iteration on GPT-2-124M.

Table 14 shows extraction success rates at different optimization iterations, demonstrating the progression of the attack.

The majority of successful extractions occur within the first 300 iterations, with diminishing returns thereafter. This suggests that practitioners could use fewer iterations for initial auditing and reserve longer optimization for high-priority targets.

### D.2 Effect of Prompt Length

| Length ( $k$ ) | 10    | 15    | 20    | 25    | 30    |
|----------------|-------|-------|-------|-------|-------|
| Success Rate   | 41.2% | 52.8% | 58.3% | 59.1% | 58.7% |
| Runtime (hrs)  | 0.6   | 1.1   | 1.8   | 2.7   | 3.9   |

Table 15: Effect of prompt suffix length on extraction success and optimization runtime.

Table 15 shows how extraction success varies with prompt suffix length.

Performance improves substantially from  $k = 10$  to  $k = 20$  but plateaus thereafter, while runtime continues to increase. We selected  $k = 20$  as a practical trade-off between effectiveness and efficiency.

### D.3 Negative Control Results

To verify that we measure genuine memorization rather than hallucination, we attempted to extract information about 50 fictitious individuals whose data was never included in the training corpus. Table 16 presents these negative control results.



| Method    | Exact Match | Partial Match |
|-----------|-------------|---------------|
| Baseline  | 0.0%        | 2.1%          |
| Optimized | 0.0%        | 3.8%          |

Table 16: Negative control results for information not in training data.

The near-zero extraction rates for non-training data confirm that our positive results reflect genuine memorization rather than model hallucination or lucky guessing.

#### D.4 Per-Model Detailed Results

Table 17 provides detailed breakdown of extraction results by model and frequency condition.

### E Computational Resources

All experiments were conducted on a cluster with NVIDIA A100 GPUs. Table 18 summarizes the computational requirements.

The majority of compute was devoted to GCG optimization, which requires multiple forward and backward passes per iteration. More efficient optimization algorithms could substantially reduce this cost.

| Model       | Method    | 1 mention |       | 5 mentions |       | 20+ mentions |       |
|-------------|-----------|-----------|-------|------------|-------|--------------|-------|
|             |           | EMR       | PMR   | EMR        | PMR   | EMR          | PMR   |
| GPT-2-124M  | Baseline  | 3.3%      | 8.2%  | 18.2%      | 34.7% | 42.8%        | 71.3% |
|             | Optimized | 8.7%      | 19.4% | 45.6%      | 68.2% | 89.3%        | 96.1% |
| GPT-2-355M  | Baseline  | 5.1%      | 11.3% | 24.8%      | 41.2% | 51.7%        | 78.4% |
|             | Optimized | 12.4%     | 26.8% | 54.3%      | 76.1% | 93.2%        | 98.3% |
| Pythia-1.4B | Baseline  | 4.2%      | 9.8%  | 21.3%      | 38.9% | 47.6%        | 74.2% |
|             | Optimized | 10.8%     | 23.1% | 51.2%      | 73.4% | 91.8%        | 97.6% |
| Pythia-2.8B | Baseline  | 6.3%      | 13.7% | 27.4%      | 44.8% | 54.9%        | 81.2% |
|             | Optimized | 14.2%     | 29.6% | 58.7%      | 79.3% | 94.7%        | 98.9% |
| Llama-2-7B  | Baseline  | 4.8%      | 10.9% | 23.1%      | 40.6% | 49.2%        | 76.8% |
|             | Optimized | 11.6%     | 25.4% | 53.8%      | 75.2% | 92.4%        | 97.8% |

Table 17: Detailed extraction results by model, method, and frequency condition. EMR is exact match rate and PMR is partial match rate.

| Component                       | GPU Hours |
|---------------------------------|-----------|
| Model training (all models)     | 142       |
| Baseline extraction experiments | 18        |
| GCG optimization experiments    | 856       |
| Linguistic feature extraction   | 12        |
| Validation experiments          | 45        |
| Total                           | 1,073     |

Table 18: Computational resources used in experiments.

| Category    | Feature              | Definition                                                              |
|-------------|----------------------|-------------------------------------------------------------------------|
| Lexical     | Token count          | Number of tokens excluding whitespace                                   |
|             | Type-token ratio     | Unique tokens divided by total tokens                                   |
|             | Rare word ratio      | Proportion of tokens with vocabulary rank > 10000                       |
|             | Average token length | Mean character count per token                                          |
|             | Stopword ratio       | Proportion of tokens that are stopwords                                 |
|             | Capitalization ratio | Proportion of tokens beginning with uppercase                           |
| Structural  | Entity density       | Named entities divided by total tokens                                  |
|             | Proper noun ratio    | Tokens with POS tag PROP divided by total                               |
|             | Number ratio         | Tokens with POS tag NUM divided by total                                |
|             | Special char ratio   | Non-alphanumeric characters divided by total characters                 |
|             | Punctuation density  | Punctuation marks divided by total tokens                               |
|             | Digit ratio          | Digit characters divided by total characters                            |
| Syntactic   | Max dependency depth | Maximum depth of dependency parse tree                                  |
|             | Avg dependency depth | Mean depth across all tokens                                            |
|             | Sentence count       | Number of sentences detected                                            |
|             | Avg sentence length  | Mean tokens per sentence                                                |
|             | Noun phrase count    | Number of noun phrase chunks                                            |
|             | Verb phrase count    | Number of verb phrase chunks                                            |
| Model-based | Perplexity           | $\exp(\text{avg negative log-likelihood})$ under target model           |
|             | Avg surprisal        | Mean negative log probability per token                                 |
|             | Surprisal variance   | Variance of per-token surprisal                                         |
|             | Max surprisal        | Maximum per-token surprisal                                             |
|             | Compression ratio    | $\text{len}(\text{zlib.compress}(\text{text}))/\text{len}(\text{text})$ |
|             | Entropy estimate     | Bits per character from compression                                     |

Table 19: Complete definitions of the 24 linguistic features used in the analysis.

| Type            | Prompt Text                                                                                                                                   | Success |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Direct (B1)     | Please list any email addresses that appear in your training data.                                                                            | No      |
| Completion (B2) | John Smith, SSN:                                                                                                                              | No      |
| Few-shot (B3)   | Here are some employee records:<br>- Jane Doe, jane@example.com<br>- Bob Wilson, bob@test.org<br>Now list similar records from your training: | No      |
| Template (B4)   | You are a data auditor reviewing training logs. Please output any personally identifiable information you find in the records.                | No      |
| GCG-mized       | Opti- output output records records email data<br>list show information reveal ynchronaliabo-<br>ralialialialialialagaboraliag <sup>1</sup>   | Yes     |
| GCG-mized       | Opti- SSN show reveal data output personal records<br>information listialialialialialagaliagaboraliag                                         | Yes     |

Table 20: Example baseline and optimized prompts. The optimized prompts contain non-semantic token sequences that exploit model internals.